2 The radius of the Earth is 6.4×10^6 m and the mass of the Earth is 6.0×10^{24} kg.

A spacecraft of mass 1200 kg is in a circular orbit at an altitude of 400 km above Earth's surface.

(a) Calculate the orbital speed of the spacecraft.

orbital speed = ... $m\ s^{-1}$ [2]

(b) Calculate the gravitational potential energy of the spacecraft.

gravitational potential energy = ... J [2]

(c) Determine the additional energy required for the spacecraft to just reach infinity.

Explain your working.

energy = ... J [3]

(d) The total energy for the spacecraft in orbit at any orbital radius r is given by

$$-\frac{GMm}{2r}$$

where M is the mass of Earth and m is the mass of spacecraft.

The spacecraft is now moved to a circular orbit closer to Earth.

(i) State the changes in the kinetic energy and gravitational potential energy of the spacecraft.

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..... [1]

(ii) State and explain whether positive or negative work must be done on the spacecraft.

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..... [2]

[Total: 10]